

HUIWON JANG

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RESEARCH INTERESTS

My research interests lie in representation learning and robotic foundation models, along with their applications. I previously focused on visual representation learning, aiming to perceive visual content and understand the world effectively and efficiently. Recently, I have been particularly focused on developing generalist robot policies. As part of this, I led the development of RLDX-1, a vision-language-action model (VLA) for generalizable and dexterous robotic manipulation.

My current work focuses on representation learning for video model-based robot policies (i.e., World Action Models), including improving the representations learned by video diffusion models and leveraging human video data to learn representations that transfer from human behavior to robotic manipulation. I submitted recent work in these directions to ICLR 2027.

EDUCATION

Mar. 2023 – Present	Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea PhD in Artificial Intelligence Advisor: Prof. Jinwoo Shin
Mar. 2019 – Feb. 2023	Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea B.S. in Mathematical Science & Computer Science (Double Major)

PUBLICATIONS

T: Technical Report, **C**: Conference, **P**: Preprint, **W**: Workshop, *,[§]: Equal contribution

[T1] RLDX-1 Technical Report

Huiwon Jang^{*,§}, Dongyoung Kim^{*,§}, Myungkyu Koo^{*}, Suhyeok Jang^{*}, Taeyoung Kim^{*}, Beomjun Kim, Byungjun Yoon, Chang-sung Jang, Daewon Choi, Dongsu Han, Donguk Lee, Heeseung Kwon, Hojin Jeon, Jaehyun Kang, Jaekyoung Bae, Jihyuk Lee, Jimin Lee, John Won, Joonwoo Ahn, Junhyeong Park, Junyoung Sung, Kyungmin Lee, Minseong Han, Minsung Yoon, SeJune Joo, Seonil Son, Seungcheol Park, Seunggeun Cho, Seungjun Moon, Seungku Kim, Yonghoon Dong, Yongjin Cho, Youngchan Kim, Chang Hwan Kim, Dohyeon Kim, Heecheol Kim, Heewon Lee, Hensen Ahn, Hyungkyu Ryu, Hyunsoo Choi, Hyunsoo Shin, Jaeheon Jung, Jaewoo Kim, Jinwook Kim, Joochul Chang, Joonsoo Kim, Junghun Park, Jungwoo Park, Junho Cho, Junhyeok Park, Junwon Lee, Kangwook Lee, Kwanghoon Kim, Kyoungwhan Choe, Manoj Bhadu, Nayoung Oh, Sangjun Kim, Sangwoo Kim, Seunghoon Shim, Seunghyun Kim, Seungjun Lee, Seungyup Ka, Sungryol Yang, Wook Jung, Yashu Shukla, Yeonjae Lee, Yeonwoo Bae, Jinwoo Shin[§]

[C10] Modular Sensory Stream for Integrating Physical Feedback in Vision-Language-Action Models

Jimin Lee, **Huiwon Jang**, Myungkyu Koo, Jungwoo Park, Jinwoo Shin

- Conference on Robot Learnin (CoRL), 2026

[P3] RoboAlign: Learning Test-Time Reasoning for Language-Action Alignment in Vision-Language-Action Models

Dongyoung Kim, Sumin Park, Woomin Song, Seungku Kim, Taeyoung Kim, **Huiwon Jang**, Jinwoo Shin, Jaehyung Kim^{*}, Younggyo Seo^{*}

[C9] SpatialBoost: Enhancing Visual Representation through Language-guided Reasoning

Byungwoo Jeon^{*}, Dongyoung Kim^{*}, **Huiwon Jang**, Insoo Kim, Jinwoo Shin

- European Conference on Computer Vision (ECCV), 2026
- Neural Information Processing Systems Workshop on Space in Vision, Language, and Embodied AI (NeurIPSW-SpaVLE), 2025

[C9] Dual-Stream Diffusion for World-model Augmented Vision-Language-Action Model

John Won, Kyungmin Lee, **Huiwon Jang**, Dongyoung Kim, Jinwoo Shin

- International Conference on Machine Learning (ICML), 2026

[P2] ContextVLA: Vision-Language-Action Model with Amortized Multi-Frame Context

Huiwon Jang, Sihyun Yu, Heeseung Kwon, Hojin Jeon, Younggyo Seo*, Jinwoo Shin*

[C8] Robot-R1: Reinforcement Learning for Enhanced Embodied Reasoning in Robotics

Dongyoung Kim, Sumin Park, **Huiwon Jang**, Jinwoo Shin, Jaehyung Kim*, Younggyo Seo*

- Conference on Neural Information Processing Systems (NeurIPS), 2025

[C7] Efficient Long Video Tokenization via Coordinate-based Patch Reconstruction

Huiwon Jang, Sihyun Yu, Jinwoo Shin, Pieter Abbeel, Younggyo Seo

- IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025

[C6] Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think

Sihyun Yu, Sangkyung Kwak, **Huiwon Jang**, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, Saining Xie

- International Conference on Learning Representations (ICLR), *Oral Presentation*, 2025

[C5] TrackIME: Enhanced Video Point Tracking via Instance Motion Estimation

Seong Hyeon Park, **Huiwon Jang**, Byungwoo Jeon, Sukmin Yun, Paul Hongsuck Seo, Jinwoo Shin

- Conference on Neural Information Processing Systems (NeurIPS), *Spotlight Presentation*, 2024

[C4] Adversarial Robustification via Text-to-Image Diffusion Models

Daewon Choi*, Jongheon Jeong*, **Huiwon Jang**, Jinwoo Shin

- European Conference on Computer Vision (ECCV), *Oral Presentation*, 2024

[C3] Visual Representation Learning with Stochastic Frame Prediction

Huiwon Jang, Dongyoung Kim, Junsu Kim, Jinwoo Shin, Pieter Abbeel, Younggyo Seo

- International Conference on Machine Learning (ICML), 2024

[C2] Modality-agnostic Self-supervised Learning with Meta-learned Masked Auto-encoder

Huiwon Jang*, Jihoon Tack*, Daewon Choi, Jongheon Jeong, Jinwoo Shin

- Conference on Neural Information Processing Systems (NeurIPS), 2023

[C1] Unsupervised Meta-learning via Few-shot Pseudo-supervised Contrastive Learning

Huiwon Jang*, Hankook Lee*, Jinwoo Shin

- International Conference on Learning Representations (ICLR), *Spotlight presentation*, 2023
- Neural Information Processing Systems Workshop on Meta-Learning (NeurIPSW-MetaLearn), 2022

[P1] AltUB: Alternating Training Method to Update Base Distribution of Normalizing Flow for Anomaly Detection

Yeongmin Kim*, **Huiwon Jang***, Dongkeon Lee, Hojin Choi

- Arxiv preprint, 2022

EXPERIENCES

Visiting Research Fellow

RLWRLD

- Co-leading research on RLDX-1

Seoul, Korea

Jun. 2025 – Jun. 2026

Research Intern

i-SENS, CGMS Algorithm Team

- Time-series regression, and forecasting
- Self-supervised learning for time-series

Seoul, Korea

Oct. 2022 – Oct. 2023

Undergraduate Research Intern

Algorithmic Intelligence Laboratory (ALIN-LAB), KAIST AI.

Advisor: [Prof. Jinwoo Shin](#)

Daejeon, Korea

Jun. 2021 – Feb. 2023

- Unsupervised meta-learning via self-supervised learning for few-shot image classification ([C1])

AWARDS & HONORS

Travel Awards, CVPR 2025 Jun. 2025

Travel Awards, ICML 2024 Jul. 2024

Travel Awards, NeurIPS 2023 Dec. 2023

Travel Awards, ICLR 2023 May. 2023

Travel Awards, Google Conference Scholarships (APAC) May. 2023

INVITED TALKS

Modality-agnostic Self-supervised Learning with Meta-learned Masked Auto-encoder

Samsung Electronics Device Solution (DS) Suwon, Korea. Oct. 2024

Samsung AI Forum (SAIF) 2023 Suwon, Korea. Nov. 2023

Samsung Advanced Institute of Technology (SAIT) Suwon, Korea. Jun. 2023

Unsupervised Meta-learning via Few-shot Pseudo-supervised Contrastive Learning

International Conference on Learning Representations (ICLR) Kigali, Rwanda. May. 2023